F18CE2 - MV Calculus and Real Analysis B: Solutions 7

1 Rolle's Theorem: Let f be continuous on $[a,b]$ and differentiable on (a,b). If $f(a) = f(b)$ then there exists $c \in (a,b)$ such that $f'(c) = 0$.

Let $f(x) = x^5 - 5\,x + \gamma$. Suppose that there are two solutions, say $a < b$, so that

$$f(a) = f(b) = 0.$$

Then by Rolle's Theorem, $f'(c) = 0$ for some c with $c > a \geqslant 1$. However, we have

$$f'(x) = 5x^4 - 5 = 5(x^4 - 1) > 0, \quad \text{for } x > 1,$$

which contradicts the previous sentence. Hence, the equation $x^5 - 5\,x + \gamma = 0$ cannot have two solutions greater than 1
(NB this does not tell us anything about whether it has any solutions, i.e., whether it has zero or one solution).

2 Define $f(x) = x + e^x - 2$. Then f is differentiable and continuous on $\mathbb{R}$.

We first show that there is a solution of $f(x) = 0$. We have

$$f(0) = -1, \quad f(2) = e^2 > 0,$$

so by the Intermediate Value theorem we have $f(c) = 0$ for some $c \in (0,2)$.

Now suppose that $f(a) = f(b) = 0$, for some a, $b \in \mathbb{R}$, with $a < b$.

Then by Rolle's Theorem, $f'(c) = 0$ for some $c \in (a,b)$. But $f'(x) = 1 + e^x > 0$ for all $x \in \mathbb{R}$. This is a contradiction so two solutions of $f(x) = 0$ cannot exist.

This completes the proof, but just to summarize what we have shown:

(a) there is at least one solution of the equation, and it is in $[-1,2]$;

(b) there is at most one solution anywhere on $\mathbb{R}$
(the numbers a, b and x in the second part of the solution can be anywhere in $\mathbb{R}$, not just in $[-1,2]$).

Combining (a) and (b) shows that there is exactly one solution in $\mathbb{R}$ (and we now also know that it is in $[-1,2]$, although the question did not ask that).

3 Define $f(x) = x^5 + x^3 + x - 1$. Then f is differentiable and continuous on $\mathbb{R}$ and

$$f(-1) = -4, \qquad f(1) = 2.$$

Hence by the Intermediate Value theorem we have $f(c) = 0$ for some $c \in (-1,1)$.

Now suppose that there are two solutions, that is, $f(a) = f(b) = 0$ for some a, b, with $a < b$.

By Rolle's Theorem, $f'(d) = 0$ for some $d \in (a,b)$. Also, $f'(x) = 5x^4 + 3x^2 + 1 > 0$ for all x.

This is a contradiction so two (or more) solutions of $f(x) = 0$ cannot exist.

4 Define $f(x) = xe^{\sin x} - \cos x$. Then f is differentiable and continuous on $\mathbb{R}$ and

$$f(0) = -1, \qquad f(\pi/2) = \frac{\pi e}{2} > 0.$$

Hence by the Intermediate Value theorem we have $f(c) = 0$ for some $c \in (0, \pi/2)$. That is, there is at least one solution.

Now suppose that $f(a) = f(b) = 0$ with $0 < a < b < \pi/2$.

By Rolle's Theorem, $f'(d) = 0$ for some $d \in (a, b)$. Also,

$$f'(x) = e^{\sin x} + x \cos x \, e^{\sin x} + \sin x.$$

For $x \in (0, \pi/2)$, $\sin x > 0$, $\cos x > 0$ so that $f'(x) > 0$.

This is a contradiction so two solutions of $f(x) = 0$ cannot exist.

5 We use the method described in the question.

(a) Define $f(x) = x + \cos x - 1$. Then,

$$f(0) = 0 + 1 - 1 = 0 \quad \text{and} \quad f'(x) = 1 - \sin x \geqslant 0 \quad \text{for all } x \geqslant 0.$$

(b) Define $f(x) = \ln(1 + x) - x + \dfrac{x^2}{2}$. Then

$$f(0) = 0 \quad \text{and} \quad f'(x) = \frac{1}{1+x} - 1 + x = \frac{x^2}{1+x} \geqslant 0 \quad \text{for all } x \geqslant 0.$$

(c) Define $f(x) = (1 + x)^\alpha - 1 - x^\alpha$. Then

$$f(0) = 0 \quad \text{and} \quad f'(x) = \alpha(1+x)^{\alpha-1} - \alpha x^{\alpha-1} \geqslant 0 \quad \text{for all } x \geqslant 0.$$

This proves the first inequality in the question.
To prove the second inequality, put $x = a/b$ in the first inequality and multiply by b^α.

(d) Define $f(x) = (1 + x)^n - 1 - nx$. Then

$$f(0) = 0 \quad \text{and} \quad f'(x) = n(1+x)^{n-1} - n \geqslant 0 \quad \text{for all } x \geqslant 0.$$

(e) Define $f(x) = (1 + x)^n - 1 - nx - \frac{1}{2}n(n-1)x^2$. Here we use the second MVT. We have

$$\begin{aligned} f(x) &= (1+x)^n - 1 - nx - \frac{1}{2}n(n-1)x^2, & f(0) &= 0, \\ f'(x) &= n(1+x)^{n-1} - n - n(n-1)x, & f'(0) &= 0, \\ f''(x) &= n(n-1)(1+x)^{n-2} - n(n-1), & f'(x) &\geqslant 0 \quad (\text{since } n \geqslant 1). \end{aligned}$$

So, by the second MVT, $f(x) \geqslant 0$, for $x > 0$, which proves the result.

6 (a) Following the hint gives,

$$1 + b_n = 2^{1/n} \iff 2 = (1+b_n)^n \geqslant 1 + nb_n \implies 0 \leqslant b_n \leqslant \frac{1}{n} \implies \lim_{n\to\infty} b_n = 0.$$

(b) Following method in part (a) gives,

$$1+b_n = n^{1/n} \iff n = (1+b_n)^n \geqslant 1 + nb_n + \frac{1}{2}n(n-1)b_n^2 \geqslant \frac{1}{2}n(n-1)b_n^2$$
$$\implies 0 \leqslant b_n^2 \leqslant \frac{2}{n-1} \implies \lim_{n\to\infty} b_n = 0.$$

Note that if we only used the first mean value theorem here we would not get convergence – at one point we needed an $n(n-1)$ term on the right hand side to cancel off an n on the left. We did not need this in part (a).

7 Let $f(x) = x\ln x$. Then f is continuous on $[r, r+1]$ and differentiable on $(r, r+1)$. Applying the MVT to f on the interval $[r, r+1]$ gives

$$f(r+1) - f(r) = f'(c)$$

for some $c \in (r, r+1)$, that is

$$f(r+1) - f(r) = 1 + \ln c \leqslant 1 + \ln(r+1).$$

Summing this from $r = 1$ to $r = n-1$ gives

$$\sum_{r=1}^{n-1} [f(r+1) - f(r)] \leqslant \sum_{r=1}^{n-1} [1 + \ln(r+1)]\,. \tag{1}$$

The LHS of (1) is

$$\sum_{r=1}^{n-1} [f(r+1) - f(r)] = [f(2) - f(1)] + [f(3) - f(2)] + [f(4) - f(3)] + \cdots + [f(n) - f(n-1)],$$
$$= f(n) - f(1) = f(n),$$

while the RHS of (1) is

$$\sum_{r=1}^{n-1} [1 + \ln(r+1)] = n - 1 + \ln n!$$

so it follows from (1) that

$$n\ln n \leqslant n - 1 + \ln n!$$

8 Let $f(x) = \sqrt{x}$. Applying the MVT to f on the interval $[n, n+1]$ shows that there exists $c \in (n, n+1)$ such that

$$\sqrt{n+1} - \sqrt{n} = f'(c) = \frac{1}{2}c^{-1/2}. \tag{2}$$

Since $n < c < n+1$, we have

$$(n+1)^{-1/2} < c^{-1/2} < n^{-1/2},$$

and substituting these inequalities into (2) gives

$$\frac{1}{2}\,(n+1)^{-1/2} \leqslant \sqrt{n+1} - \sqrt{n} \leqslant \frac{1}{2}\,n^{-1/2}\,.$$

Now, using this and the definition of a_n gives

$$a_n - a_{n+1} = 2(n+1)^{1/2} - 2n^{1/2} - (n+1)^{-1/2} \geqslant 0.$$

Therefore (a_n) is decreasing.